

Asfiya Misba

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EDUCATION

Master of Science in Computer Science (GPA: 4.0/4.0)

The University of Texas at Arlington

Specialization: Intelligent Systems, Software Engineering

2024

Arlington, TX

Bachelor of Engineering in Computer Science (GPA: 9.6/10.0)

RNS Institute of Technology, affiliated to Visvesvaraya Technological University

2022

Bengaluru, India

SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP, C++
- **Web & Mobile Development:** Spring Boot, React.js, Node.js, HTML, CSS, Django, React Native, Android Studio, Git, VS Code, Postman
- **Databases:** MySQL, Oracle, PostgreSQL
- **Cloud & DevOps:** AWS (EC2, S3), UTA Cloud, Docker, REST APIs
- **Tools & Methodologies:** Agile Methodology, Unit Testing (Mocha, JUnit), Swagger, Tableau, Microsoft Office (Excel, PowerPoint, Word)

PROFESSIONAL EXPERIENCE

Research Assistant, The University of Texas at Arlington

Aug 2024 - Present

- Developed a Java Spring Boot web app with Leaflet.js to visualize simulated signal strength and network coverage maps.
- Adapted large language models (LLMs) for telecom-specific tasks to improve network optimization and domain relevance.
- Built a signal-aware pathfinding model using OpenStreetMap and Dijkstra's algorithm to compute optimal connectivity routes.
- Contributed to a C++ traffic light control system using state-machine logic to simulate realistic urban traffic behavior.

Graduate Research Assistant, The University of Texas at Arlington

Sep 2023 - May 2024

- Revamped OER web interfaces using HTML, CSS, JavaScript, and React, improving accessibility and user experience.
- Created interactive content with H5P to boost engagement across educational platforms.
- Curated and evaluated open resources for the Digital Information Literacy Hub on OERTX, increasing content relevance and reach.

Student Associate, The University of Texas at Arlington

Jun 2023 - Aug 2023

- Utilized LaTeX, HTML, CSS, and JavaScript to support content development and customization, resulting in improvement in the overall learning experience and customization of learning materials.
- Analyzed textbook savings data in Tableau, identifying patterns that demonstrated a 75% cost reduction for students.

Data Science Intern, LocalHost

Sep 2021 - Dec 2021

- Built a predictive salary model with 81% accuracy using Pandas, NumPy, and Seaborn.
- Participated in a flight price prediction hackathon, delivering a model with 72% efficiency.
- Performed data cleaning and feature engineering, ensuring high-quality input for model training.

ACADEMIC PROJECTS

HealthMate App

- Developed HealthMate, a cross-platform fitness mobile app enabling fitness goal tracking features like step counting and a water intake reminder using React Native with a Node.js/MySQL backend and Redux for state management, enabling seamless tracking and reminders.
- Conducted manual and automated testing with Mocha, ensuring a reliable and user-friendly experience.

Learning Management System

- Built a role-based Learning Management System using HTML, CSS, PHP, and MySQL, integrating authentication, access control, and communication features.
- Deployed on the UTA cloud for scalable performance and accessibility.

Book Rental System

- Built a Book Rental System using Java, Spring Boot, and MySQL with RESTful APIs for user, book, and rental management.
- Implemented JWT-based authentication with Spring Security to enable secure, role-based access.
- Designed a layered backend architecture and tested endpoints using Postman for reliability and REST compliance.

Transaction Management

- Implemented a C/C++ transaction manager using S2PL for concurrency control, ensuring serializability.
- Designed deadlock prevention and optimized lock granularity for high-concurrency performance.

ACHIEVEMENTS

- Recipient of the Gold Star Scholarship at The University of Texas at Arlington (UTA) for exceptional academic achievement.
- Earned the Maverick Advantage Distinction at UTA, recognizing leadership, research contributions, and professional development.
- Awarded 6th Rank by Visvesvaraya Technological University (VTU) for outstanding academic performance in my Bachelor's degree.
- Received the Gold Medal from RNS Institute of Technology (RNSIT) in recognition of my academic excellence during my undergraduate studies.
- Co-authored and presented two journal papers on CNN-based age and gender detection (2022) and AI safety (2021) at the NCCT-ICT national conference.